Iso Lab (EKS Spot, L2) replay of the priced DP/eq residual index-calculus study, p36

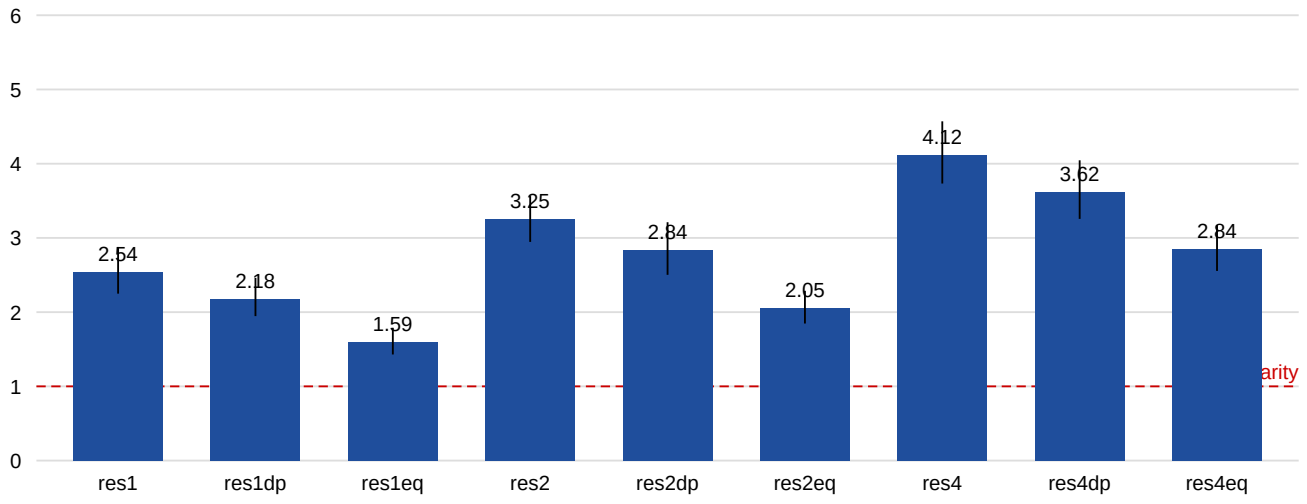
Class: infrastructure verification. Not a speedup. The goal (index calculus near rho parity) is not met: the best arm, res1eq, is 1.591 × matched rho at w=0 (ratio of pooled means, the harness's estimator), 1.13 × above its √2 collision floor.

Outcomes

- **R1 determinism.** All 528 records common to the local L0 run (rounds 1–3 × 16 workloads × 11 arms) have identical counted GAE, recovered scalar and algorithm seed (11 arms × 48 = 528/528).
- **R2 admission gates.** Session complete; audit.json ok, 0 problems, 1056 records; 880 replays, all `identical`; L2 gate accepted all 1056 executions (80/80 pairs at L2); A/A res2 vs res2aa wall ratio 0.9995 [0.9981, 1.0017] (within [0.9, 1.1]); quality score 85. SHA256SUMS: 19/19 OK. Source commit be464efac844aa20598a769a085820d2c7101ae9 (+ musl ioctl fix), ecbench sha256 2ce2d100…ed95, image digest sha256:dcb51a3a…83e0bb, Job UID 4c53b06b-5201-47f7-9dc2-116df3697d21.
- **Harness comparison res2aa/rho** (priced w=0/0.1/1): 3.2491 / 3.2489 / 3.2478, CI95 [2.918, 3.617] over 80 pairs on 16 workloads. Wall ns/GAE ratio 1.4629 [1.442, 1.484], outside the A/A noise: the IC arm costs ~46% more wall time per counted operation than rho (memory-bound lookups). It is an upper bound on cost per operation, not a verdict.

Ratios to matched rho (w=0)

Ratio of pooled mean counted GAE over 5 rounds × 16 workloads (the harness's estimator; reproduces res2aa/rho = 3.249); bootstrap over workloads, 4000 resamples. Last column: ratio divided by the collision floor √(B+1) (B = 1, 2, 4 for res1, res2, res4). An earlier draft used a mean of per-workload ratios and reported res1eq as 2.11 ×; that estimator is skewed by rho's per-workload variance and is withdrawn. Priced columns (w=0.1, 1) come only from the harness comparison; the IC records carry no uncharged counters.



ratio of pooled mean counted GAE to matched rho, w=0, p36, Iso Lab L2, 5 rounds x 16 workloads; whiskers = 95% workload bootstrap

arm	ratio	95% CI	/ floor
res1	2.542	[2.249, 2.878]	1.80
res1dp	2.175	[1.946, 2.462]	1.54
res1eq	1.591	[1.430, 1.789]	1.13
res2	3.249	[2.946, 3.563]	1.88
res2dp	2.838	[2.503, 3.211]	1.64
res2eq	2.055	[1.847, 2.287]	1.19
res4	4.116	[3.733, 4.571]	1.84
res4dp	3.623	[3.257, 4.046]	1.62

res4eq	2.844	[2.555, 3.174]	1.27

Caveats

- Same unpriced residual-table-bytes counter caveat as the parent study (validator F1); owner decision outstanding.
- Pre-existing validation gate: no benchmarks/README row until VALIDATION.json covers artifact-store raw data.
- One curve (p36); rounds 4–5 are new data with no L0 counterpart.

Evidence: s3 launcher receipt and artifacts of run 41342132360; protocol research/ic_residual_dp_eq_isolab_20261101/PROTOCOL.md (Amendments 1–2).